SOMISETTY VENKATA SURENDRA KUMAR

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Professional Summary

Cloud Engineer Trainee with 0.8 years of hands-on experience in building and automating cloud infrastructure and CI/CD pipelines. Proficient in AWS services, Docker containerization, and leveraging infrastructure-as-code principles to ensure operational readiness and reliability. Driven by a problem-solving mindset, I contribute to stable, efficient, and quality-minded platform operations.

Education

Siddharth Institute of Engineering and Technology *2023-09 – 2026-01*

B.Tech - [GPA: 8.2]

Work Experience

Cloud Engineer Trainee - *[Q Spiders]* *2026-01 - Present*

- Engineered end-to-end Jenkins CI/CD pipelines with Maven, SonarQube quality gates, and Nexus artifact storage, ensuring consistent code quality and streamlined deployments.
- Standardized container platform deployments using Docker, accelerating application rollouts and establishing reusable platform patterns for efficient runtime operations.
- Provisioned and administered core AWS infrastructure (EC2, VPC, IAM, S3, Lambda) for automated deployments, ensuring foundational stability and security with least-privilege access.
- Implemented monitoring and alerting solutions using AWS CloudWatch and SNS, enhancing operational visibility and automating recurring tasks via Bash scripts for improved efficiency.

Projects

CI/CD Deployment Automation for Java Web Application

- Designed a fully automated Jenkins pipeline that takes a Java web application from Git commit to live deployment with zero manual release steps.
- Configured GitHub webhooks to trigger builds automatically on every push, ensuring each code change is built and validated within minutes.
- Streamlined pipeline stages: source checkout, Maven compile/test/package, SonarQube code-quality analysis, artifact publishing to Nexus, Docker image build, and containerized deployment.
- Standardized the release process so every deployment is repeatable, versioned, and traceable to a specific commit.
- **Technologies:** Jenkins, Maven, Docker, GitHub

S3 Event-Driven Alarm & Notification System

- Built an event-driven monitoring solution that replaced manual log checking with automated incident detection on Amazon S3 activity.
- Created CloudWatch alarms on S3 activity and error thresholds to detect abnormal behavior in real-time.
- Integrated Amazon SNS topics and subscriptions to push real-time notifications on every alarm state change, reducing time to detect and respond to issues.
- **Technologies:** Amazon S3, CloudWatch, SNS

Dockerized Web Application Deployment on AWS

- Wrote Dockerfiles and built custom, versioned Docker images for a web application, publishing them to Docker Hub.
- Configured robust container networking, ensuring reliable communication between application components.
- Deployed and operated the application on an AWS EC2 Linux instance by pulling versioned images from Docker Hub, enabling consistent, repeatable deployments and simple rollbacks.
- **Technologies:** Docker, Docker Hub, AWS EC2, Linux

Skills

- **Technical**: Linux administration, DNS, HTTP/HTTPS, load balancing, subnetting, CIDR, NAT, infrastructure-as-code, pipeline-as-code, image builds, container networking, relational database design, SQL (SELECT, JOINs, GROUP BY, subqueries, aggregate functions, DDL/DML), observability, high-availability
- **Tools**: Kubernetes, Docker, AWS (EC2, VPC, IAM, S3, EBS, RDS, Lambda, Elastic Load Balancing, Auto Scaling, Route 53, CloudFront, CloudWatch, SNS), Jenkins, GitHub webhooks, Maven, SonarQube, Nexus, Git, GitHub, Bash, Shell scripting, Apache Tomcat, Amazon RDS

Certifications

- **Operating System Fundamentals - [NPTEL24CS108S650802155]** *2024-10*
  NPTEL

Awards

- Best Paper Award, ICICC-2026 (International Conference on Innovative Computing and Communication), for "HealthGuard: A Collaborative Machine Learning Approach to Secure Medical Information Across IoT-Driven Healthcare Systems."

Publications

- Published research paper in Springer: link.springer.com/chapter/10.1007/978-3-032-30008-9_35